

First Terminal Examination - 2079

Class : XI

Subject: Physics (1011)

Full Marks: 75

Time: 3 hrs.

SET 'A'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group A

Circle the correct one from given alternatives.

[11 × 1 = 11]

1. Which of the following is dimensional formula for specific heat capacity?
 - a. $[M^0L^2T^{-2}K^{-1}]$
 - b. $[M^0L^2T^{-3}K^{-1}]$
 - c. $[M^0LT^{-2}K^{-1}]$
 - d. $[ML^2T^{-2}K^{-1}]$
2. The number of significant figures in 7316000 are
 - a. 7
 - b. 6
 - c. 5
 - d. 4
3. A vernier caliper of least count 0.1mm is used to measure the length of tube. The length of tube is found to be 5cm long. How should this measurement be recorded in a table of results?
 - a. 5 cm
 - b. 5.0 cm
 - c. 5.00 cm
 - d. 5.000 cm
4. Which of the following pair have same dimensions?
 - a. Force and torque
 - b. Planck's constant and linear momentum
 - c. Force and potential energy
 - d. Torque and potential energy

5. A force F is given by $F = at + bt^2$ where t is time. What is the dimension of b ?
- $[MLT^{-4}]$
 - $[MLT^{-3}]$
 - $[ML^2T^{-4}]$
 - $[MLT^4]$
6. A metal sheet has a circular hole in it. When it is heated the diameter of the hole
- increases
 - decreases
 - does not change
 - may increase or decrease
7. In case of concave mirror, the minimum distance between a real object and its real image is
- f
 - $2f$
 - $4f$
 - zero
8. A virtual image larger than the object can be produced by
- convex mirror
 - concave mirror
 - plane mirror
 - both a and b
9. Two plane mirrors are inclined at 90° with each other. The number of images formed is
- 2
 - 3
 - 4
 - 5
10. A shaving mirror of focal length ' f ' produces an image ' x ' times the size of the object. If the image is real, then the distance of the object from the mirror is:
- $(x+1)f$
 - $(x+1)f/x$
 - $(x-1)f/x$
 - $(x-1)f$
11. What is the magnification when the object is placed at $2f$ from the pole of a convex mirror?
- $-1/3$
 - $-2/3$
 - -1
 - $-3/2$

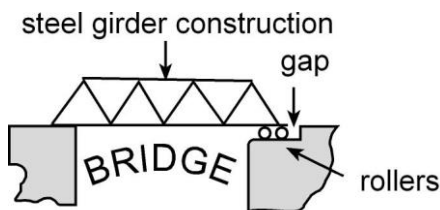
Group – B

Give short answer to the following questions.

[8 × 5 = 40]

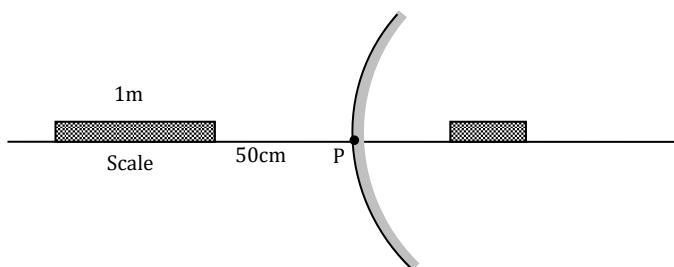
12. a. Check the dimensional correctness of the equation $T = 2\pi\sqrt{\frac{m}{K}}$, where
‘T’ is time period, ‘m’ is mass and ‘K’ is restoring force per unit extension. [2]
- b. The force ‘F’ is given in terms of time ‘t’ and displacement ‘x’ by the equation $F = A \sin Bx + C \sin Dt$. What are the dimensions of D/B and A? [2]
- c. State the principle of homogeneity of dimension. [1]
13. a. Using the method of dimension, show that the viscous force ‘F’ acting on sphere of radius ‘r’ moving with a terminal velocity ‘v’ in liquid of coefficient of viscosity ‘ η ’ is $F=6\pi\eta rv$. [3]
- b. Convert the density of water 1 g/cm³ into kg/m³ using dimensional analysis. [2]
14. a. Obtain the dimensional formula for refractive index and potential energy. [2]
- b. The length of rod is exactly 5cm. An observer records the reading as 5.0cm, 5.00cm and 5.000cm. Which is the most accurate measurement and why? [2]
- c. Differentiate between accuracy and precision in measurement. [1]
15. a. Suppose you have a piece of wood and a piece of iron. [2]
- i. What is their minimum possible temperature?
- ii. At what temperature will wood and iron appear equally hot or equally cold to our body?
- b. A Centigrade and a Fahrenheit thermometer are placed in hot water. The water is then cooled. What fall of temperature will the Fahrenheit thermometer register, when the Centigrade thermometer records the fall in temperature as 45°C? [3]

16. Consider this diagram of a bridge over a river.



- a. Briefly explain why you think one end is constructed on rollers. [1]
 - b. When would you expect the gap indicated in the diagram to be biggest - in the winter time or the summer time? [2]
 - c. What, do you think, would happen if the bridge had been constructed with no gap? (Give an answer for construction in midwinter and midsummer) [2]
17. a. Establish the relation between focal length and radius of curvature in case of concave mirror. [2]
- b. What change in focal length is observed if the convex mirror is dipped in water? [1]
 - c. A concave mirror forms a magnified image of a real object. What are the two possible positions of the object in front of the mirror? Explain. [2]
18. a. An object is situated 40 cm from a convex mirror. When a plane mirror is inserted between object and convex mirror at a distance of 32 cm from the object, the images in the mirrors coincide. What is the focal length of convex mirror? [3]
- b. What type of mirror will you suggest for shaving or make-up purpose and why? [2]

19. a. Is mirror formula applicable to plane mirror? Explain. [2]
- b. A meter - scale is placed along the axis of convex mirror focal length 25 cm, its nearer end being at a distance of 50 cm shown in figure. Calculate size of the image formed. [3]



Group – C

Give long answer to the following questions.

[3 × 8 = 24]

20. a. What are dimensions? Using the method of dimension show that the centripetal force 'F' acting on object of mass 'm' moving with uniform speed 'v' in a circle of radius 'r' is $F = \frac{mv^2}{r}$. [1+3]
- b. Whether there are physical quantities having unit but no dimension, illustrate with example? Is its reverse possible? Explain. [2]
- c. What are significant figures? Find the number of significant figures in the measurements 151.00 cm and $0.05 \times 10^2 \text{m}$. [2]
21. a. We are lucky that the earth isn't in thermal equilibrium with the sun which has a surface temperature of about 5800K. But, why aren't they in thermal equilibrium? [2]

- b. What is sensitivity of thermometer? Two thermometers are constructed in the same way except that one has a spherical bulb and the other an elongated cylindrical bulb. Which one will respond quickly to the temperature? Explain [1+2]
- c. A thermometer has wrong calibration. It reads the melting point of ice as -10°C . It reads 60°C in place of 50°C . What is the temperature of boiling point of water on this scale? [3]
22. a. Derive the mirror formula for concave mirror when virtual image is formed. [3]
- b. What is the difference between virtual image formed by convex and concave mirror? [1]
- c. Can a convex mirror ever form a real image? Show in figure. [1]
- d. A mirror forms an erect image 30 cm from the object and twice its height. [3]
- i. Where must the mirror be situated?
- ii. What is the radius of curvature?
- iii. Assuming the object to be real, determine whether the mirror is convex or concave.

ॐ - The End - ॐ